Project Smart Trigger for ECE/CS 3992

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***Abstract*—This project proposes a smart trigger for firearms that uses RFID technology to authenticate the user and ensure that only authorized users can fire the gun. This would help to prevent accidental shootings and unauthorized use of firearms, ultimately improving public safety. This paper will discuss solutions already in the market, the proposed solution, and the ethical considerations that need to be taken into account when designing and implementing such a system. It will also explore the project and outline the tasks that need to be completed in order to successfully implement the smart trigger and demonstrate its functionality at Demo Day. The risks associated with the project will also be assessed and strategies for mitigating those risks will be discussed.**

I. Problem Statement

A huge problem with firearms is that they are dangerous and can cause serious harm or death if not handled properly. This is especially true when it comes to children getting ahold of firearms or unauthorized users who may not know how to handle a firearm safely. Police officers can lose their firearms in the line of duty and if they are not recovered they can be used in crime. The proposed system, a smart trigger, would help to address this problem by preventing the gun from firing unless it is used by the owner. Use of RFID technology to authenticate the user and ensure that only authorized users can fire the gun. This would help to prevent accidental shootings and unauthorized use of firearms, ultimately improving public safety.

II. Solution Space

Efforts to improve firearm safety through access control have evolved through several distinct technological approaches. At the most basic level, traditional solutions rely on purely mechanical safeguards such as trigger locks or integrated key-based locking mechanisms. These systems are inexpensive and highly reliable because they do not depend on electronics or power, but they do not restrict access to a specific user. Anyone with access to the key or combination can operate the firearm, which causes concern toward illegitimate use. In time-sensitive or high-stress situations, the need to manually unlock a device with a key can render the safety ineffective in practice, leading many police and military personnel to avoid using such systems. This feature creates a gap between theoretical safety and real-world usage, which encouraged us to develop a system that enforces access control without requiring manual intervention.

Electronic authentication methods represent the next step in this progression. Systems based on PIN entry, external key fobs, or smartphone connectivity attempt to introduce user-specific control while maintaining relatively simple hardware implementations. While these approaches improve upon mechanical systems by enabling programmable access, they still require explicit interaction from the user, such as entering a code or activating a device. This introduces latency and cognitive load that can be unacceptable in defensive scenarios. Additionally, wireless-based systems may be vulnerable to interception or spoofing, depending on their implementation. As a result, these approaches highlight the need for authentication mechanisms that are both secure and implicit, allowing the system to verify the user seamlessly without requiring conscious action.

Biometric authentication has emerged as the most prominent alternative in this space, particularly in commercial “smart gun” designs. By leveraging physiological traits such as fingerprints, palm vein patterns, or even grip characteristics, biometric systems aim to bind firearm operation directly to the identity of the user. This approach is compelling because it eliminates the need for external tokens and ensures that only authorized individuals can operate the weapon. However, when examined in the context of real-world firearm use, biometrics introduce a number of critical challenges. Sensors such as fingerprint scanners are highly sensitive to environmental conditions. Metrics like dirt, sweat, gloves, or moisture can significantly degrade performance. Firearms are often used in unpredictable and harsh environments, where such conditions are common, leading to an increased likelihood of authentication failure.

Latency is another major concern. Biometric systems

require time to capture, process, and match data against stored templates. Even small delays can be unacceptable in scenarios where immediate response is necessary. Equally important is the issue of reliability in terms of false negatives and false positives. A false negative, where an authorized user is denied access, represents a potentially dangerous failure in a defensive situation. Conversely, a false positive undermines the entire purpose of the system. These risks are compounded by the long-term degradation of sensors and the variability of human biological signals. In addition to these technical concerns, biometric systems tend to be significantly more expensive and power-intensive than alternative solutions. Commercial implementations often cost on the order of $1000 to $1500, reflecting the complexity of most biometric devices. This cost and power burden conflicts directly with the constraints of a small, battery-powered module designed to fit within the firing control group of an AR-15.

RFID-based authentication offers a different point in the design space by prioritizing simplicity, robustness, and low power consumption. In a passive near-field RFID system, authentication occurs when a tag enters the electromagnetic field generated by a reader, allowing the tag to transmit its unique identifier without requiring its own power source. This enables extremely low-latency operation and reduces the system's overall energy requirements. RFID systems are also less sensitive to environmental contaminants compared to optical or capacitive biometric sensors, making them more reliable in varied conditions. The primary tradeoff is that RFID relies on possession rather than inherent identity. A user must carry an external token, such as a ring or card, which introduces the possibility of loss or unauthorized use if the token is compromised.

More complex hybrid systems attempt to combine multiple authentication methods, such as pairing RFID with biometrics or behavioral sensing, in order to improve security. While these approaches can reduce certain vulnerabilities, they significantly increase system complexity, cost, and power consumption. For a compact, retrofittable module operating under strict physical and electrical constraints, this hybrid complexity is impractical. As a result, the solution space reflects a series of tradeoffs between security, reliability, usability, and feasibility. Biometric systems offer strong identity assurance but struggle with robustness and cost, while RFID systems sacrifice some degree of identity binding in exchange for speed, reliability, and practicality. This balance makes RFID a practical choice for a system intended to be compact, cost-effective, and deployable in real-world conditions.

III. Related Work

Radio frequency identification (RFID) has been widely studied as part of the broader class of automatic identification technologies. Kaur et al. provides a comprehensive overview of its principles, capabilities, and limitations. RFID systems enable the wireless transmission of identification using radio waves, allowing individuals or objects to be identified without requiring line-of-sight interaction, in contrast to traditional barcode systems [3]. This capability significantly improves automation and reduces the need for human action within lock systems, making RFID useful for embedded and access control applications that require high speed and reliability.

A major focus of prior work is the distinction between passive and active RFID systems, as well as the physical mechanisms that enable their operation. Passive RFID operates by harvesting energy from the electromagnetic field generated by a reader. In near-field systems, this process is governed by magnetic induction, where an alternating magnetic field induces a voltage across a tag's coil, which provides enough energy for the tag to transmit its stored ID [3]. This eliminates the need for an onboard power source. These characteristics make passive RFID well-suited for compact, power-constrained embedded systems, such as our intended implementation within a firearm's firing control group.

Despite these advantages, Kaur et al. also identifies several important limitations of RFID systems that must be addressed. One key constraint is the relationship between operating frequency and read range. In near-field systems, the effective communication range is limited and decreases rapidly due to the inverse cubic decay of the magnetic field [3]. While this restricts long-range communication, it can be advantageous for security sensitive applications by ensuring that the authenticated user is within close proximity. Other challenges include signal collision when multiple tags are present, environmental interference caused by conductive materials or electromagnetic noise, and the lack of universal standardization across RFID systems [3]. Security and privacy concerns are also significant, as unauthorized cloning of tags can compromise system integrity if appropriate safeguards are not implemented.

Taken together, the existing literature establishes RFID as a mature and well-understood technology for identification and access control. It has demonstrated advantages in automation, robustness, and low power operation. At the same time, it highlights key technical and security challenges that must be carefully addressed in our implementation. This paper provides a strong foundation for

the proposed system, informing both the design choices and the tradeoffs involved in applying RFID to firearm safety.

IV. Solution / Contribution

To truly understand our project and our approach, an overview of the firing control group of an AR-15 rifle is necessary. The firing control group operates through a precise mechanical interaction between the trigger, sear, and hammer. As seen in Fig 1, the hammer is held back under spring tension by the sear. When the trigger is pulled, it disengages the sear, releasing the hammer and allowing it to rotate forward where it would eventually hit the firing pin.



Fig. 1. AR-15 Firing Control Group [2]

Our proposed solution is to block the firing control group by applying an additional force against the hammer, ensuring that it can never strike the firing pin. Our idea is to use a linear actuator to provide this force against the hammer. A linear actuator paired with a microcontroller would allow for low-voltage operation of our access control. The gun would then only be able to fire when a validated RFID tag is present in close proximity to our implemented system on the firing control group.

The primary contribution of this project is the integration of RFID-based access control into this already operating, small mechanical system. A passive RFID reader embedded near the control group continuously scans for a valid tag worn by the user, particularly in a ring worn on their finger. When an authorized tag is detected, the system signals the actuator to disengage, allowing normal operation of the firing mechanism. If no valid tag is present, the actuator blocks the hammer from hitting the firing pin. This way the firearm would never fire even if the trigger is pressed. This approach enables fast and implicit authentication without requiring user interaction, while also remaining low-power and cost-effective compared to biometrics alternatives. By combining a simple mechanical intervention with proximity-based RFID authentication, the system provides a practical solution for enhancing firearm safety.

V. Safety, global, societal, ethical considerations

What are the public health, safety and welfare, global, cultural, societal, environmental, and economic factors related to the problem and solution space? What power does the proposed system provide, and to whom? What are the relevant professional ethical obligations to consider in the design of the system?

A. Factors Related to the Problem and Solution

1) Public Health, Safety, and Welfare: The main issue the proposed system addresses is the potential for negligent discharge of a firearm. This is because of the nature of guns in that the are designed to inflict harm on a target. The smart trigger would help to protect the public safety in a couple of ways. First the system would be designed to prevent the gun from firing unless it is used by the owner. This would help to prevent accidental shootings by children or others that are not lisenced to shoot a firearm. Secondly, if the gun was lost unless the module was removed the gun wouldn't be able to fire. This would help to prevent the gun from being used in crime or from being used in a negligent way. Lastly, the system is modular meaning that if at any point it has a malfunction the module can be removed and the gun can be used as normal. This would help to prevent situtation where a gun is rendered unusable due to malfucntions of the system. In the past since the mid 1990s, there have been numerous starts to create something similar to the smart trigger. These safety devices are known as Firearm Safety Devices (FSDs) and are most of the time mechanical as they are reliable and don't require a power source as shown in Fig 2.

They also have some characteristics and features that are mentioned in [1]

- the option of integrating and combining mechanical devices with various authorization processes (token, password, biometric);
- hierarchical authorization;
- traceability and logging functions;
- networking and remote availability.



Fig. 2. Trigger and Cable Lock [1]

2) Global, and Cultural: Because our project is focused on firearms it is impotant to consider the global and cultural factors that may affect the implementation and use of the smart trigger. For example, different countries have different laws and regulations regarding firearms, and the cultural attitudes towards guns can vary significantly. It is important to ensure that the system is designed in a way that is compliant with these regulations and respectful of these cultural differences. For the scope of the project we will be focusing on the United States, but it is also important to consider its implications in other countries. The United States has a deep history with firearms and the Second Amendment to the Constitution guarantees the right of citizens to keep and bear arms. This doesn't mean that no regulations have been put in place regarding firearms. Some laws that may affect the system include:

- Maryland (SB 334/HB 1339) This law requires that all firearms be equipped with a safety device before being sold or transferred. In order for our system to approved in Maryland it would need to be approved by the Handgun Roster Board, which is responsible for evaluating and approving firearms and firearm safety devices for sale in the state.
- New Jersey's "Display Mandate": This law requires that all firearms be displayed in a visible manner when being sold or transferred. This would require the entire state to adopt the smart trigger as the standard to use in New Jersey.
- Because our system uses RFID technology its legaly and intential radiator. This means that if we ever wanted to sell the system in the United States we would need to get approval from the Federal Communications Commission (FCC) to ensure that it complies with regulations regarding radio frequency emissions.

3) Societal and Economic: Part of the reason why the RFID authentication would be benefictial is that it would be a more affordable and reliable solution compared to other biometric authentication methods. This would make it more accessible to a wider range of gun owners, which could help to increase the adoption of the system and ultimately improve public safety. At gun ranges unless the user was wearing the RFID tag they wouldn't be able to use the gun. This would make shooting ranges safer for everyone because it would prevent unauthorized users from using the guns. This would also help to prevent accidents and injuries at shooting ranges, which could have a positive impact on public health and safety. Another way in which the system would increase safety is if a gun was lost or stolen from the police department. If the gun was equipped with the smart trigger it would be rendered useless to any thief.

TABLE I
DEVELOPMENT COST

Microcontroller	\$15 or free
RFID Scanner	$9 prototype board$,3-5 per chip
AR-15 Lower	Free (thanks Stephen)
3D Printed Parts	\$20 maximum
Custom PCB	\$50 in prototyping, $\sim$25 per order
Parts for the PCBs	\$20-50
Linear Actuator	\$20
RFID Tags	$10 for prototypes$,10 for the ring
RFID Programmer	\$30
Battery	$10 for battery$,5 for charger parts
Total	**\$190-235**

4) Environmental: The use of RFID technology in our system also has some environmental considerations. The production and disposal of the RFID tags and other electronic components can have an impact on the environment. However, we will strive to use sustainable materials and practices wherever possible to minimize this impact.

B. Power

Understanding how the system could be used and how power is shifting is important to ensure that the system is designed in a way that is ethical and responsible. The smart trigger would provide power to the gun owner by allowing them to have more control over who can use their firearm. This would help to prevent unauthorized use of the gun, which could improve public safety and reduce the risk of accidents and injuries. This would also give power to law enforment by making it more difficult for stolen guns to be used in crimes and against police officers in shoot outs. The system would also provide power to the general public by helping to prevent accidents and injuries caused by negligent discharge of firearms. If the system were to have any bugs or issues it could potentially be exploited by moving power to the criminals.

VI. Proposed Work

A project includes some things and excludes others you know you won't work on. Define more explicitly what is included and not included here. What will be demonstrated at project completion at Demo Day?

For our project, we will include parts of the gun that are necessary to demo the smart trigger functionality. Items Included:

- Lower receiver: This is the part of the gun that contains the trigger mechanism and is necessary for demonstrating the smart trigger functionality.
- RFID reader: This is the component that will be used to authenticate the user and ensure that only authorized users can fire the gun.
- RFID tag: This is the component that will be attached to the user's ID and will be read by the RFID reader to authenticate the user.
- Microcontroller: This is the brain of the smart trigger system and will be responsible for processing the RFID data and controlling the trigger mechanism.
- Custom PCB: This will be used to connect all the components together and ensure that they work together seamlessly.

Items Excluded:

- Upper receiver: This is the part of the gun that contains the barrel and is not necessary for demonstrating the smart trigger functionality.
- Stock: This is the part of the gun that provides support and is not necessary for demonstrating the smart trigger functionality.
- Grip: This is the part of the gun that is held by the user and is not necessary for demonstrating the smart trigger functionality.

VII. Project Tasks

A. Major Milestones

- Planning:
  - Full bill of matierals made
  - Major research complete
  - Safety precautions addressed
  - Due: *03 Aug 2026*
- Functional RFID reader:
  - Able to read and log tags
  - Exportable log
  - Due: *24 Aug 2026*
- Application for updating authorized tags:
  - User able to update authorized tags without reflashing
  - Due: *04 Sep 2026*
- Successful blocking of fire control:
  - Method of blocking the firing of the AR-15 solidified and location confirmed
  - Due: *11 Sep 2026*
- Part fitting:
  - Adapting all components into the small form factor required by constraints
  - Due: *02 Oct 2026*
- Assembly:
  - All components fit together and operate in basic expected function
  - Due: *30 Oct 2026*
- Endurance testing:
  - Full system able to withstand operation of firearm
  - Due: *13 Nov 2026*

B. Testing

Testing will be done in stages. We for all software we will have a suite of unit tests to ensure that edge cases are handeled elegantly and primary function is correct. All hardware will be tested on proto-board wire ups, then in firm soldered PCB's. Each piece will need to be able to survive a decent amount of impact and final testing will include physical endurance testing to ensure the project is hardy and durable.

C. Task interfaces

RFID Tag: An RFID Tag is a small device that contains an antenna. When in the presence of a reading antenna it transmit the data stored on it. Typically this data is simple with a communication key and an ID number for the reader to compare against its authorized list. This will be the primary input device of the final product.

Solenoid: The primary output of the final product will be a linear actuating solenoid. This is an electronic device that moves a metal bar. This physical bar is what will block the fire control group of the AR-15.

User Updates: A key feature will be the logging of all tags that attempted to unlock the firearm. The project includes generating an exportable list of this log. We will also inlcude the ability for the authorized user list to be updated, and change the tags that will allow the operation of the firearm.

D. Design Approach

The design of the smart trigger system follows a modular, iterative approach to integrate mechanical, electrical, and software components into the existing AR-15 firing control group.

Hardware and PCB Design: We are using KiCad for the development of our custom Printed Circuit Board (PCB). This open-source suite allows for precise schematic capture and multi-layer PCB layout, which is necessary to meet the strict spatial constraints of the firearm's grip and receiver. For mechanical housing and the linear actuator interface, we utilize a 3D-modeling toolchain (AutoCAD, Inventor, or SolidWorks) to design parts for additive manufacturing. These tools allow us to verify tolerances before printing, ensuring the solenoid interacts correctly with the hammer and sear.

Software Development: The firmware for the system's microcontroller is developed in VS Code, utilizing an interrupt-driven state machine to manage power consumption. We will use C/C++ or MicroPython for low-level hardware abstraction, ensuring low-latency communication between the RFID reader and the MCU.

Measurement and Validation: To ensure reliability, we use digital oscilloscopes and logic analyzers to verify signal integrity between the RFID module and the MCU. Mechanical components are validated using precision calipers to ensure the 3D-printed components fit within the standard AR-15 lower receiver without interfering with existing mechanical safeties.

E. Current Status

Our team has completed the foundational research and administrative setup necessary for the project's execution. To date, the following milestones have been achieved:

- **Infrastructure and Version Control:** A shared Git repository has been established to manage firmware development and LaTeX documentation, ensuring seamless collaboration and version tracking.
- **Research and Review:** We have completed the initial Research Draft and a secondary analysis of existing market solutions, such as biometric and mechanical FSDs. These drafts underwent both peer and instructor review, leading to the solidification of our RFID-based approach.
- **Exploratory Prototyping:** We have conducted preliminary experimentation with RFID technology and STM32 microcontrollers during previous laboratory work. While we have not finalized the specific MCU or RFID chipset for the final build, this sandbox testing confirmed the feasibility of the proximity-based authentication required for the ring-tag interface.

F. Stretch Goals

While the core project is defined by successful RFID-based blocking of the firing pin, we have identified several stretch goals that would enhance the system's security and usability:

- **Mobile Configuration App:** Developing a companion Android application (potentially via Kotlin) that allows users to manage authorized tags, view access logs, and check battery levels over a secure Bluetooth link.
- **Encrypted Tag Authentication:** Moving from basic UID reading to an encrypted handshake (such as AES-128 or Mifare DESFire) to prevent tag cloning or spoofing attacks.
- **Haptic Feedback:** Integrating a small vibration motor within the grip to provide the user with tactile confirmation when the system is successfully unlocked, reducing the cognitive load during high-stress scenarios.
- **Ultra-Low Power Deep Sleep:** Implementing an even more aggressive power management profile that uses an accelerometer to "wake" the system only when the firearm is moved, further extending battery life.

VIII. Integration Strategy

Our project has a constraint to fit within a very small space. We will utilize a partially distributed integration strategy in order to mesh all of the components together. The primary electronics are an RFID module, a microcontroller unit (MCU), a power supply, and a linear actuating solenoid. All components except the solenoid will be housed in the plastic grip for an AR-15. The plastic grip has the benefit of not interfering with the RFID signal thus reducing the complexity of a faraday cage effect that could block any electromagnetic signal from being identified.

The solenoid will be housed inside the actual receiver of the AR-15. This separate location allows for the direct blocking of the fire control group. The actuation will be triggered via fine gauge wire passing through one of the holes where the grip physically attaches to the receiver.

Power management is one of the most challenging parts of this entire project. The RFID module can consume up to 60mA during its operation and the solenoid up to 200mA. To compensate for these power requirements we will employ an interrupt driven state machine. A low power timer will periodically activate the RFID module to check for a tag, if one is detected it will activate hardware interrupt on the MCU to check if the tag is authorized. If the tag is in the authorized list the MCU will then actuate the solenoid to remove the block of the fire control group, ultimately allowing the firing of the AR-15.

IX. Risk Assessment

To ensure project success and the safety of all personnel, we have identified and mitigated key risks associated with the development of a smart trigger system. Our methodology involved a comprehensive risk matrix where each hazard was initially categorized as Red (high impact/likelihood), Yellow (moderate), or Green (low). Through a proactive, iterative de-risking process, we have implemented technical and procedural safeguards to transition all identified hazards into the Green category.

- **Safety and Legal Compliance (Initially Red):** Potential for bringing a functional firearm onto a school campus; non-compliance with state/federal firearm and radio frequency regulations; risk of negligent discharges during testing.
- **Technical and Operational (Initially Yellow):** Inadequate reliability in real-world environments; mechanical failure of the linear actuator or solenoid; potential for pinch injuries from high-force moving components.
- **Design and Integration (Initially Green):** Severe spatial constraints within the fire control group; electrical interference between the RFID scanner and the metallic receiver.

Campus Safety and Legal Compliance: The most critical risk identified was the presence of a firearm on school grounds. We have fully de-risked this by utilizing a modular design that excludes the upper receiver, barrel, and firing pin from all on-campus activities. Only the smart trigger assembly and lower receiver are brought to campus for demonstration. This creates a physical impossibility of discharge while allowing for full verification of the electronic lockout. To address regulatory concerns, we are consulting with university police and safety officers to ensure compliance with local statutes. Additionally, as the RFID module is an intentional radiator, we are designing the system to adhere to FCC standards for radio frequency emissions. These combined measures transition these risks to **Green**.

Safe Handling and Negligent Discharge: To mitigate risks of improper handling, all team members have completed a safety orientation focusing on the four primary laws of firearm safety. This was particularly important as some members began the project with limited firearm experience. All initial functionality testing is conducted in a controlled lab environment without a firearm present. By using an interrupt-driven state machine to manage the 200mA solenoid actuation, we ensure the device only disengages the mechanical block when a validated RFID tag is confirmed, preventing accidental activation. These strict protocols ensure a **Green** safety rating.

Testing and Mechanical Reliability: To address the risk of component failure, we have implemented a multi-stage validation process. This includes individual stress testing of the solenoid to ensure it can withstand the spring tension of the hammer. Because power management is a primary challenge, with the RFID module and solenoid drawing significant current, we are testing the system's endurance to ensure the battery can reliably actuate the device under repeated use. Field testing in varied environments will further identify potential interference from conductive materials or electromagnetic noise. This ensures the system is robust, moving these concerns to **Green**.

Physical Hazards and Spatial Design: Potential for minor injuries from moving parts is mitigated through ergonomic housing designs and the use of personal protective equipment (PPE) during assembly. Regarding spatial constraints, the requirement to fit the MCU and power supply within the AR-15 grip is handled through high-fidelity 3D modeling and custom PCB fabrication using KiCad. We utilized precise measurements of the firing control group to ensure that the linear actuator provides the necessary force without interfering with the existing mechanical sear. By verifying these tolerances through detailed prototypes, both risks are confirmed **Green**.

With these mitigation strategies finalized and integrated into our development cycle, the project is fully de-risked and prepared for a safe demonstration.

X. Conclusion

This project proposes an innovative solution to firearm safety. By utilizing near-field RFID technology, the system provides users with a secure firearm that remains immediately accessible when needed. We are developing a modular access-control solution for the AR-15 platform to enhance safety and prevent unauthorized use. Our primary goal is to promote the safe handling and operation of firearms by integrating a seamless, electronic authorization step into the gun's mechanical function. A fully functional prototype will be completed and rigorously tested for presentation at Senior Demo Day.

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